SECTION 087100
DOOR HARDWARE

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Mechanical and electrified door hardware for:
   a. Swinging doors.

2. Field verification, preparation and modification of existing doors and frames to receive new door hardware.
3. The intent of the hardware specification is to specify the hardware for interior and exterior doors, and to establish a type, continuity, and standard of quality. However, it is the contractor’s responsibility to thoroughly review existing conditions, schedules, specifications, drawings, and other Contract Documents to verify the suitability of the hardware specified.

B. Section excludes:

1. Windows
2. Cabinets (casework), including locks in cabinets
3. Signage
4. Toilet accessories
5. Overhead doors
6. Shower doors
7. Access doors and panels
8. Conduit, junction boxes and wiring.

C. Related Sections:

1. Division 01 Section "Alternates" for alternates affecting this section.
2. Division 06 Section "Rough Carpentry"
3. Division 06 Section "Finish Carpentry"
4. Division 07 Section "Joint Sealants" for sealant requirements applicable to threshold installation specified in this section.
5. Division 08 Sections:
   a. "Metal Doors and Frames"
   b. "Flush Wood Doors"
   c. "Stile and Rail Wood Doors"
   d. "Interior Aluminum Doors and Frames"
   e. "Aluminum-Framed Entrances and Storefronts"
   f. "Stainless Steel Doors and Frames"
   g. "Special Function Doors"
   h. "Entrances"

6. Division 09 sections for touchup, finishing or refinishing of existing openings modified by this section.

7. Division 26 "Electrical" sections for connections to electrical power system and for low-voltage wiring.
8. Division 28 "Electronic Safety and Security" sections for coordination with other components of electronic access control system and fire alarm system.

1.2 REFERENCES

A. UL LLC

1. UL 10B - Fire Test of Door Assemblies
2. UL 10C - Positive Pressure Test of Fire Door Assemblies
3. UL 1784 - Air Leakage Tests of Door Assemblies
4. UL 305 - Panic Hardware

B. DHI - Door and Hardware Institute

1. Sequence and Format for the Hardware Schedule
2. Recommended Locations for Builders Hardware
3. Keying Systems and Nomenclature
4. Installation Guide for Doors and Hardware

C. NFPA – National Fire Protection Association

1. NFPA 70 – National Electric Code
2. NFPA 80 – 2019 Edition – Standard for Fire Doors and Other Opening Protectives
3. NFPA 101 – Life Safety Code
4. NFPA 105 – Smoke and Draft Control Door Assemblies
5. NFPA 252 – Fire Tests of Door Assemblies

D. ANSI - American National Standards Institute

1. ANSI A117.1 – 2017 Edition – Accessible and Usable Buildings and Facilities
2. ANSI/BHMA A156.1 - A156.29, and ANSI/BHMA A156.31 - Standards for Hardware and Specialties
3. ANSI/BHMA A156.28 - Recommended Practices for Keying Systems
4. ANSI/WDMA I.S. 1A - Interior Architectural Wood Flush Doors
5. ANSI/SDI A250.8 - Standard Steel Doors and Frames

E. WHI – Warnock Hersey Incorporated

F. WI – Woodwork Institute

G. AWI – Architectural Woodwork Institute

H. NAAMM – National Association of Architectural Metal Manufacturers

I. Local applicable codes

1. Use date of standard in effect as of Bid Date.

1.3 SUBMITTALS

A. General:

1. Submit in accordance with Conditions of Contract and Division 01 Submittal Procedures.
2. Prior to forwarding submittal:
   a. Review drawings and Sections from related trades to verify compatibility with specified hardware.
   b. Highlight, encircle, or otherwise specifically identify on submittals: deviations from Contract Documents, issues of incompatibility or other issues which may detrimentally affect the Work.
   c. Prior to forwarding submittal, comply with procedures for verifying existing door and frame compatibility for new hardware, as specified in PART 3, “EXAMINATION” article, herein.

B. Action Submittals:

1. Product Data: Submit technical product data for each item of door hardware, installation instructions, maintenance of operating parts and finish, and other information necessary to show compliance with requirements.
2. Riser and Wiring Diagrams: After final approval of hardware schedule, submit details of electrified door hardware, indicating:
   a. Wiring Diagrams: For power, signal, and control wiring and including:
      1) Details of interface of electrified door hardware and building safety and security systems.
      2) Schematic diagram of systems that interface with electrified door hardware.
      3) Point-to-point wiring.
      4) Risers.

3. Samples for Verification: If requested by Architect, submit production sample of requested door hardware unit in finish indicated and tagged with full description for coordination with schedule.
   a. Samples will be returned to supplier. Units that are acceptable to Architect may, after final check of operations, be incorporated into Work, within limitations of key coordination requirements.

4. Door Hardware Schedule:
   a. Submit concurrent with submissions of Product Data, Samples, and Shop Drawings. Coordinate submission of door hardware schedule with scheduling requirements of other work to facilitate fabrication of other work critical in Project construction schedule.
   b. Submit under direct supervision of a Door Hardware Institute (DHI) certified Architectural Hardware Consultant (AHC) or Door Hardware Consultant (DHC) with hardware sets in vertical format as illustrated by Sequence of Format for the Hardware Schedule published by DHI.
   c. Indicate complete designations of each item required for each opening, include:
      1) Door Index: door number, heading number, and Architect's hardware set number.
      2) Quantity, type, style, function, size, and finish of each hardware item.
      3) Name and manufacturer of each item.
      4) Fastenings and other pertinent information.
      5) Location of each hardware set cross-referenced to indications on Drawings.
      6) Explanation of all abbreviations, symbols, and codes contained in schedule.
      7) Mounting locations for hardware.
      8) Door and frame sizes and materials.
      9) Degree of door swing and handing.

10) Operational Description of openings with any electrified hardware (locks, exits, electromagnetic locks, electric strikes, automatic operators, door position switches, magnetic holders or closer/holder units, and access control components). Operational description should include operational descriptions for: egress, ingress (access), and fire/smoke alarm connections.

d. Submittal Sequence: Submit door hardware schedule concurrent with submissions of Product Data, Samples, and Shop Drawings. Coordinate submission of door hardware schedule with scheduling requirements of other work to facilitate fabrication of other work that is critical in Project construction schedule.

5. Key Schedule:
   a. After Keying Conference, provide keying schedule that includes levels of keying, explanations of key system's function, key symbols used, and door numbers controlled.
   b. Use ANSI/BHMA A156.28 "Recommended Practices for Keying Systems" as guideline for nomenclature, definitions, and approach for selecting optimal keying system.
   c. Provide 3 copies of keying schedule for review prepared and detailed in accordance with referenced DHI publication. Include schematic keying diagram and index each key to unique door designations.
   d. Index keying schedule by door number, keyset, hardware heading number, cross keying instructions, and special key stamping instructions.
   e. Provide one complete bitting list of key cuts and one key system schematic illustrating system usage and expansion. Forward bitting list, key cuts and key system schematic directly to Owner, by means as directed by Owner.
   f. Prepare key schedule by or under supervision of supplier, detailing Owner's final keying instructions for locks.

6. Templates: After final approval of hardware schedule, provide templates for doors, frames and other work specified to be factory or shop prepared for door hardware installation.

C. Informational Submittals:

1. Provide Qualification Data for Supplier, Installer and Architectural Hardware Consultant.
2. Provide Product Data:
   a. Certify that door hardware approved for use on types and sizes of labeled fire-rated doors complies with listed fire-rated door assemblies.
   b. Include warranties for specified door hardware.

3. Certificates of Compliance:
   a. UL listings for fire-rated hardware and installation instructions if requested by Architect or Authority Having Jurisdiction.
   b. Installer Training Meeting Certification: Letter of compliance, signed by Contractor, attesting to completion of installer training meeting specified in “QUALITY ASSURANCE” article, herein.
   c. Electrified Hardware Coordination Conference Certification: Letter of compliance, signed by Contractor, attesting to completion of electrified hardware coordination conference, specified in “QUALITY ASSURANCE” article, herein.

D. Closeout Submittals:

1. Operations and Maintenance Data: Provide in accordance with Division 01 and include:
   a. Complete information on care, maintenance, and adjustment; data on repair and replacement parts, and information on preservation of finishes.

b. Catalog pages for each product.
c. Factory order acknowledgement numbers (for warranty and service)
d. Final approved hardware schedule edited to reflect conditions as installed.
e. Final keying schedule
f. Copies of floor plans with keying nomenclature
g. Copy of warranties including appropriate reference numbers for manufacturers to identify project.
h. As-installed wiring diagrams for each opening connected to power, both low voltage and 110 volts.

E. Inspection and Testing:

1. Submit a written report of the results of functional testing and inspection for fire door assemblies, in compliance with NFPA 80.
   a. Written report to be provided to the Owner and be made available to the Authority Having Jurisdiction (AHJ).
   b. Report to include the door number for each fire door assembly, door location, door and frame material, fire rating, and summary of deficiencies.

2. Submit a written report of the results of functional testing and inspection for required egress door assemblies, in compliance with NFPA 101.
   a. Written report to be provided to the Owner and be made available to the Authority Having Jurisdiction (AHJ).
   b. Report to include the door number for each required egress door assembly, door location, door and frame material, fire rating, and summary of deficiencies.

1.4 QUALITY ASSURANCE

A. If discrepancy between drawings and scheduled material in this section, bid the more expensive of the two choices, note the discrepancy in the submittal and request direction from Architect for resolution.

B. Qualifications and Responsibilities:

1. Supplier: Recognized architectural hardware supplier with a minimum of 5 years documented experience supplying both mechanical and electromechanical door hardware similar in quantity, type, and quality to that indicated for this Project. Supplier to be recognized as a factory direct distributor by the manufacturer of the primary materials with a warehousing facility in the Project’s vicinity. Supplier to have on staff, a certified Architectural Hardware Consultant (AHC) or Door Hardware Consultant (DHC) available to Owner, Architect, and Contractor, at reasonable times during the Work for consultation.
   a. Warehousing Facilities: In Project's vicinity.
   b. Scheduling Responsibility: Preparation of door hardware and keying schedules.
   c. Engineering Responsibility: Preparation of data for electrified door hardware, including Shop Drawings, based on testing and engineering analysis of manufacturer's standard units in assemblies like those indicated for this Project.
   d. Coordination Responsibility: Assist in coordinating installation of electronic security hardware with Architect and electrical engineers and provide installation and technical data to Architect and other related subcontractors.
      1) Upon completion of electronic security hardware installation, inspect and verify that all components are working properly.

2. Installer: Qualified tradesperson skilled in the application of commercial grade hardware with experience installing door hardware similar in quantity, type, and quality as indicated for this Project.
3. Architectural Hardware Consultant: Person who is experienced in providing consulting services for door hardware installations that are comparable in material, design, and extent to that indicated for this Project and meets these requirements:
   a. For door hardware: DHI certified AHC or DHC.
   b. Can provide installation and technical data to Architect and other related subcontractors.
   c. Can inspect and verify components are in working order upon completion of installation.
   d. Capable of producing wiring diagram and coordinating installation of electrified hardware with Architect and electrical engineers.

4. Single Source Responsibility: Obtain each type of door hardware from single manufacturer.
   a. Bid and submit manufacturer's updated/improved item if scheduled item is discontinued.

C. Certifications:

1. Fire-Rated Door Openings:
   a. Provide door hardware for fire-rated openings that complies with NFPA 80 and requirements of authorities having jurisdiction.
   b. Provide only items of door hardware that are listed products tested by UL LLC, Intertek Testing Services, or other testing and inspecting organizations acceptable to authorities having jurisdiction for use on types and sizes of doors indicated, based on testing at positive pressure and according to NFPA 252 or UL 10C and in compliance with requirements of fire-rated door and door frame labels.

2. Smoke and Draft Control Door Assemblies:
   a. Provide door hardware that meets requirements of assemblies tested according to UL 1784 and installed in compliance with NFPA 105
   b. Comply with the maximum air leakage of 0.3 cfm/sq. ft. (3 cu. m per minute/sq. m) at tested pressure differential of 0.3-inch wg (75 Pa) of water.

3. Electrified Door Hardware
   a. Listed and labeled as defined in NFPA 70, Article 100, by testing agency acceptable to authorities having jurisdiction.

4. Accessibility Requirements:
   a. Comply with governing accessibility regulations cited in "REFERENCES" article 087100, 1.02.D3 herein for door hardware on doors in an accessible route. This project must comply with all Federal Americans with Disability Act regulations and all Local Accessibility Regulations.

D. Pre-Installation Meetings

1. Keying Conference
   a. Incorporate keying conference decisions into final keying schedule after reviewing door hardware keying system including:
      1) Function of building, flow of traffic, purpose of each area, degree of security required, and plans for future expansion.
      2) Preliminary key system schematic diagram.

3) Requirements for key control system.
4) Requirements for access control.
5) Address for delivery of keys.

2. Pre-installation Conference
   a. Review and finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
   b. Inspect and discuss preparatory work performed by other trades.
   c. Inspect and discuss electrical roughing-in for electrified door hardware.
   d. Review sequence of operation for each type of electrified door hardware.
   e. Review required testing, inspecting, and certifying procedures.

3. Coordination Conferences
   a. Installation Coordination Conference: Prior to hardware installation, schedule and hold meeting to review questions or concerns related to proper installation and adjustment of door hardware.
      1) Attendees: Door hardware supplier, door hardware installer, Contractor.
      2) After meeting, provide letter of compliance to Architect, indicating when meeting was held and who was in attendance.
   b. Electrified Hardware Coordination Conference: Prior to ordering electrified hardware, schedule and hold meeting to coordinate door hardware with security, electrical, doors and frames, and other related suppliers.
      1) Attendees: electrified door hardware supplier, doors and frames supplier, electrified door hardware installer, electrical subcontractor, Owner, Owner’s security consultant, Architect and Contractor.
      2) After meeting, provide letter of compliance to Architect, indicating when coordination conference was held and who was in attendance.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Delivery:

1. Inventory door hardware on receipt and provide secure lock-up for hardware delivered to Project site.
2. Tag each item or package separately with identification coordinated with final door hardware schedule, and include installation instructions, templates, and necessary fasteners with each item or package. Deliver each article of hardware in manufacturer's original packaging.
3. Deliver each article of hardware in manufacturer's original packaging.

B. Project Conditions:

1. Maintain manufacturer-recommended environmental conditions throughout storage and installation periods.
2. Control handling and installation of hardware items so that completion of Work will not be delayed by hardware losses both before and after installation.

C. Protection and Damage:

1. Promptly replace products damaged during shipping.
2. Handle hardware in manner to avoid damage, marring, or scratching. Correct, replace or repair products damaged during Work.

3. Protect products against malfunction due to paint, solvent, cleanser, or any chemical agent.

D. Deliver keys to manufacturer of key control system for subsequent delivery to Owner.

1.6 COORDINATION

A. Coordinate layout and installation of floor-recessed door hardware with floor construction. Cast anchoring inserts into concrete.

B. Installation Templates: Distribute for doors, frames, and other work specified to be factory or shop prepared. Check Shop Drawings of other work to confirm that adequate provisions are made for locating and installing door hardware to comply with indicated requirements.

C. Security: Coordinate installation of door hardware, keying, and access control with Owner's security consultant.

D. Electrical System Roughing-In: Coordinate layout and installation of electrified door hardware with connections to power supplies and building safety and security systems.

E. Existing Openings:

1. Where existing doors, frames and/or hardware are to remain, field verify existing functions, conditions and preparations and coordinate to suit opening conditions and to provide proper door operation. If a conflict arises between the specified/scheduled hardware and existing conditions, then submit request for direction from Architect. Include date of jobsite visit in the submittal.
2. Submittals prepared without thorough jobsite visit by qualified hardware expert will be rejected as non-compliant.

F. Direct shipments not permitted, unless approved by Contractor.

1.7 WARRANTY

A. Manufacturer's standard form in which manufacturer agrees to repair or replace components of door hardware that fail in materials or workmanship within published warranty period.

1. Warranty does not cover damage or faulty operation due to improper installation, improper use or abuse.
2. Warranty Period: Beginning from date of Substantial Completion, for durations indicated in manufacturer's published listings.
    a. Mechanical Warranty
        1) Locks
            a) Schlage L, Keys and Cylinders: 10 years
        2) Exit Devices
            a) Von Duprin 98/99 and 33/35A: 10 years
        3) Electric Strikes
            a) Von Duprin - Mechanical components: 5 years
        4) Closers
            a) LCN 4000 Series: 30 years
        5) Overhead Stops

a) Glynn-Johnson: 10 years
6) Automatic Operators
a) LCN: 2 years
7) Weatherstrip, Seals, Thresholds
a) Zero: 5 years
b. Electrical Warranty
1) Locks
a) Schlage L, Keys and Cylinders: 3 years
2) Exit Devices
a) Von Duprin 98/99 and 33/35A: 3 year
3) Electric Strikes
a) Von Duprin - Electro-Mechanical components: 3 years
4) Power Supplies
a) Schlage and Von Duprin: 3 years

1.8 MAINTENANCE

A. Furnish complete set of special tools required for maintenance and adjustment of hardware, including changing of cylinders.

B. Turn over unused materials to Owner for maintenance purposes.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. The Owner requires use of certain products for their unique characteristics and project suitability to ensure continuity of existing and future performance and maintenance standards. After investigating available product offerings, the Awarding Authority has elected to prepare proprietary specifications. These products are specified with the notation: "No Substitute."

1. Where "No Substitute" is noted, submittals and substitution requests for other products will not be considered.

B. Approval of manufacturers and/or products other than those listed as “Scheduled Manufacturer” or “Basis of Design” shall be in accordance with substitution procedure in division 01 25 00. In the individual article for the product category items, shall be in accordance with the QUALITY ASSURANCE article, herein

C. Approval of products is contingent upon those products providing all functions and features and meeting all requirements of scheduled manufacturer's product.

D. Where specified hardware is not adaptable to finished shape or size of members requiring hardware, furnish suitable types having same operation and quality as type specified, subject to Architect's approval in accordance with substitution procedure in division 01 25 00.

2.2 MATERIALS

A. Fasteners

1. Provide door hardware manufactured to comply with published templates generally prepared for machine screw installation.
2. Furnish screws for installation with each hardware item. Finish exposed (exposed under any condition) screws to match hardware finish, or, if exposed in surfaces of other work, to match finish of this other work including prepared for paint surfaces to receive painted finish.
3. Finish exposed screws to match hardware finish, or, if exposed in surfaces of other work, to match finish of this other work including prepared for paint surfaces to receive painted finish.
4. Provide concealed fasteners for hardware units exposed when door is closed except when no standard units of type specified are available with concealed fasteners. Do not use thru-bolts for installation where bolt head or nut on opposite face is exposed in other work unless thru-bolts are required to fasten hardware securely. Review door specification and advise Architect if thru bolts are required.
   a. Coordinate with “Metal Doors and Frames”, “Flush Wood Doors”, “Stile and Rail Wood Doors” to ensure proper reinforcements.

B. Modification and Preparation of Existing Doors: Where existing door hardware is indicated to be removed and reinstalled.

1. Provide necessary fillers, Dutchmen, reinforcements, and fasteners, compatible with existing materials, as required for mounting new opening hardware and to cover existing door and frame preparations.
2. Use materials which match materials of adjacent modified areas.
3. When modifying existing fire-rated openings, provide materials permitted by NFPA 80 as required to maintain fire-rating.

C. Provide screws, bolts, expansion shields, drop plates and other devices necessary for hardware installation.

1. Where fasteners are exposed to view: Finish to match adjacent door hardware material.

D. Cable and Connectors:

1. Where scheduled in the hardware sets, provide each item of electrified hardware and wire harnesses with number and gage of wires enough to accommodate electric function of specified hardware.
2. Provide Molex connectors that plug directly into connectors from harnesses, electric locking and power transfer devices.
3. Provide through-door wire harness for each electrified locking device installed in a door and wire harness for each electrified hinge, electrified continuous hinge, electrified pivot, and electric power transfer for connection to power supplies.

2.3 HINGES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Ives 5BB series

2. Acceptable Manufacturers and Products:

a. No Substitute

B. Requirements:

1. Provide hinges conforming to ANSI/BHMA A156.1.
2. Provide five knuckle, ball bearing hinges.
3. 1-3/4 inch (44 mm) thick doors, up to and including 36 inches (914 mm) wide:
   a. Exterior: Standard weight, bronze or stainless steel, 4-1/2 inches (114 mm) high
   b. Interior: Standard weight, steel, 4-1/2 inches (114 mm) high
   c. Interior: Heavy weight steel, all doors with closers and/or overhead stops

4. 1-3/4 inch (44 mm) thick doors over 36 inches (914 mm) wide:
   a. Exterior: Heavy weight, bronze or stainless steel, 5 inches (127 mm) high
   b. Interior: Heavy weight, steel, 5 inches (127 mm) high

5. 2 inches or thicker doors:
   a. Exterior: Heavy weight, bronze or stainless steel, 5 inches (127 mm) high
   b. Interior: Heavy weight, steel, 5 inches (127 mm) high

6. Adjust hinge width for door, frame, and wall conditions to allow proper degree of opening.
7. Provide three hinges per door leaf for doors 90 inches (2286 mm) or less in height, and one additional hinge for each 30 inches (762 mm) of additional door height.
8. Where new hinges are specified for existing doors or existing frames, provide new hinges of identical size to hinge preparation present in existing door or existing frame.
9. Hinge Pins: Except as otherwise indicated, provide hinge pins as follows:
   a. Steel Hinges: Steel pins
   b. Non-Ferrous Hinges: Stainless steel pins
   c. Out-Swinging Exterior Doors: Non-removable pins
   d. Out-Swinging Interior Lockable Doors: Non-removable pins
   e. Interior Non-lockable Doors: Non-rising pins

2.4 ELECTRIC POWER TRANSFER

A. Manufacturers:

1. Scheduled Manufacturer and Product:
   a. Von Duprin EPT-10

2. Acceptable Manufacturers and Products:
   a. No Substitute

B. Requirements:

1. Provide power transfer with electrified options as scheduled in the hardware sets. Provide with number and gage of wires enough to accommodate electric function of specified hardware.
2. Locate electric power transfer per manufacturer's template and UL requirements, unless interference with operation of door or other hardware items.

2.5 PIVOT SETS

A. Manufacturers:

1. Scheduled Manufacturer:
    a. Ives

2. Acceptable Manufacturers:
    a. No Substitute

B. Requirements:

1. Provide pivot sets complete with oil-impregnated top pivot, unless indicated otherwise.
2. Where offset pivots are specified, Provide one intermediate pivot for doors less than 91 inches (2311 mm) high and one additional intermediate pivot per leaf for each additional 30 inches (762 mm) in height or fraction thereof. Intermediate pivots spaced equally not less than 25 inches (635 mm) or not more than 35 inches (889 mm) on center, for doors over 121 inches (3073 mm) high.
3. Provide appropriate model where pivot sets are scheduled at fire rated openings.

2.6 FLUSH BOLTS

A. Manufacturers:

1. Scheduled Manufacturer:
    a. Ives

2. Acceptable Manufacturers:
    a. No Substitute

B. Requirements:

1. Provide automatic, constant latching, and manual flush bolts with forged bronze or stainless-steel face plates, extruded brass levers. Provide dust-proof strikes at each bottom flush bolt.

2.7 COORDINATORS

A. Manufacturers:

1. Scheduled Manufacturer:
    a. Ives

2. Acceptable Manufacturers:
    a. No Substitute

B. Requirements:

1. Where pairs of doors are equipped with automatic flush bolts, an astragal, or other hardware that requires synchronized closing of the doors, provide bar-type coordinating device, surface applied to underside of stop at frame head.
2. Provide filler bar of correct length for unit to span entire width of opening, and appropriate brackets for parallel arm door closers, surface vertical rod exit device strikes, or other stop mounted hardware. Factory-prepared coordinators for vertical rod devices as specified.

2.8 MORTISE LOCKS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Schlage L9000 series

2. Acceptable Manufacturers and Products:
   a. No Substitute

B. Requirements:

1. Provide mortise locks conforming to ANSI/BHMA A156.13 Series 1000, Grade 1, and UL Listed for 3-hour fire doors.
2. Indicators: Where specified, provide indicator window measuring a minimum 2-inch x 1/2 inch with 180-degree visibility. Provide messages color-coded with full text and/or symbols, as scheduled, for easy visibility.
3. Provide locks manufactured from heavy gauge steel, containing components of steel with a zinc dichromate plating for corrosion resistance.
4. Provide lock case that is multi-function and field reversible for handing without opening case. Cylinders: Refer to “KEYING” article, herein.
5. Provide locks with standard 2-3/4 inches (70 mm) backset with full 3/4 inch (19 mm) throw stainless steel mechanical anti-friction latchbolt. Provide deadbolt with full 1-inch (25 mm) throw, constructed of stainless steel.
6. Provide standard ASA strikes unless extended lip strikes are necessary to protect trim.
7. Provide electrified options as scheduled in the hardware sets. Where scheduled, provide switches and sensors integrated into the locks and latches. Provide motor based electrified and motor based latch retraction locksets that comply with the following requirements:
   a. Universal input voltage – single chassis accepts 12 or 24VDC to allow for changes in the field without changing lock chassis.
   b. Fail Safe/Fail Secure – changing mode between electrically locked (fail safe) and electrically unlocked (fail secure) is field selectable without opening the lock case
   c. Low maximum current draw – maximum 0.4 amps (Lever control) and maximum 2.0 amps (Latch retraction) to allow for multiple locks on a single power supply.
   d. Low holding current (Lever control or latch retraction) – maximum 0.01 amps to produce minimal heat, eliminate “hot levers” in electrically locked applications and motorized latch retraction applications, and to provide reliable operation in wood doors that provide minimal ventilation and air flow.
   e. Connections – provide quick-connect Molex system standard.

8. Lever Trim: Solid brass, bronze, or stainless steel, cast or forged in design specified, with wrought roses and external lever spring cages. Provide thru-bolted levers with 2-piece spindles.
   a. Lever Design: 03A.

2.9 EXIT DEVICES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Von Duprin 99/33A series

2. Acceptable Manufacturers and Products:
   a. No Substitute

B. Requirements:

1. Provide exit devices tested to ANSI/BHMA A156.3 Grade 1 and UL listed for Panic Exit or Fire Exit Hardware.
2. Cylinders: Refer to "KEYING" article, herein.
3. Provide grooved touchpad type exit devices, fabricated of brass, bronze, stainless steel, or aluminum, plated to standard architectural finishes to match balance of door hardware.
4. Touchpad must extend a minimum of one half of door width. No plastic inserts are allowed in touchpads.
5. Provide exit devices with deadlatching feature for security and for future addition of alarm kits and/or other electrified requirements.
6. Provide exit devices with weather resistant components that can withstand harsh conditions of various climates and corrosive cleaners used in outdoor pool environments.
7. Provide flush end caps for exit devices.
8. Provide exit devices with manufacturer's approved strikes.
9. Provide exit devices cut to door width and height. Install exit devices at height recommended by exit device manufacturer, allowable by governing building codes, and approved by Architect.
10. Mount mechanism case flush on face of doors or provide spacers to fill gaps behind devices. Where glass trim or molding projects off face of door, provide glass bead kits.
11. Provide cylinder or hex-key dogging as specified at non fire-rated openings.
12. Removable Mullions: 2 inches (51 mm) x 3 inches (76 mm) steel tube. Where scheduled as keyed removable mullion, provide type that can be removed by use of a keyed cylinder, which is self-locking when re-installed.
13. Provide factory drilled weep holes for exit devices used in full exterior application, highly corrosive areas, and where noted in hardware sets.
14. Provide electrified options as scheduled.
15. Top latch mounting: double- or single-tab mount for steel doors, face mount for aluminum doors eliminating requirement of tabs, and double tab mount for wood doors.
16. Provide exit devices with optional trim designs to match other lever and pull designs used on the project.
17. Special Options:
    a. SI
       1) Provide dogging indicators for visible indication of dogging status.
    b. QM
       1) Rim Exit Devices: provide devices with damper-controlled re-latching to reduce operational noise. Where lever trim is specified, provide damper controlled lever return.

2.10 ACCESS CONTROL READER - FOR REFERENCE ONLY - TO BE FURNISHED, INSTALLED AND COMMISSIONED BY DIVISION 28

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Schlage MTB Series

2. Acceptable Manufacturers and Products:
   a. No Substitute

B. Requirements:

1. Provide access control card readers manufactured by a global company who is a recognized leader in the production of access control devices. Card reader manufactured for non-access control applications are not acceptable.
2. Provide multi-technology contactless readers complying with ISO 14443.
3. Provide access control card readers capable of reading the following technologies:
   a. CSN - DESFire® CSN, HID iCLASS® CSN, Inside Contactless PicoTag® CSN, ST Microelectronics® CSN, Texas Instruments Tag-It®, CSN, Phillips I-Code® CSN
   b. 125 KHz proximity - Schlage® Proximity, HID® Proximity, GE/CASI® Proximity, AWID® Proximity, LenelProx®
   c. 13.56 MHz Smart card - Schlage smart cards using MIFARE Classic® EV1/EV3, Schlage smart cards using MIFARE Plus®, Schlage smart cards using MIFARE® DESFire® EV1/EV3, Schlage smart cards using MIFARE® DESFire® EV2/EV3
   d. 13.56 MHz NFC (mobile), 2.45 GHz Bluetooth (mobile) - Mobile means compatible with Bluetooth and NFC-enabled smartphones.

2.11 ACCESS CONTROL PLATFORM

A. Manufacturers and Products:

1. Scheduled Manufacturer:
   a. Schlage Engage Commercial

2. Acceptable Manufacturers and Products:
   a. No Substitute

B. Requirements:

1. Provide a cloud-based platform capable of managing users, credentials, access rights, schedules, and audits.
2. All locks must be supplied in construction mode.
3. Provide a platform that supports a mobile application (app). Mobile application must allow for:
   a. Commissioning and configuring devices
   b. Immediately updating door files
   c. Retrieving audit information
   d. Performing firmware updates

4. Provide software set up on the owner's workstation and Mobile Device which includes:
   a. Creation of the Owner's Account
   b. Creation of the Project Site
   c. Creation of the Team as directed by the Owner
   d. Addition of five users
   e. Set up of MT20W and update firmware
   f. Create unique credentials and verify proper commissioning of ten locks

5. Provide, at the owner's request, the following on-site training prior to the expiration of the service agreement:
   a. Completing the following with ENGAGE software:
      1) Modifying the Team
      2) Move in/move out procedure including
         a) Adding and Deleting Users

b) Adding and Deleting Doors
3) Adding, assigning and programming credentials for access
4) Replacing or deleting lost credentials.
5) Retrieving and viewing of audit information
6) Assigning temporary access
b. Commissioning and verifying proper functioning between locks and credentials.
c. Updating firmware on the locks.

6. Must include a service agreement ending a year after Substantial Completion. This service agreement includes being on-site up to 16 hours for set-up and training, as listed above.

2.12 ELECTRIC STRIKES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Von Duprin 6000 Series

2. Acceptable Manufacturers and Products:
   a. No Substitute

B. Requirements:

1. Provide electric strikes designed for use with type of locks shown at each opening.
2. Provide electric strikes UL Listed as burglary resistant that are tested to a minimum endurance test of 1,000,000 cycles.
3. Where required, provide electric strikes UL Listed for fire doors and frames.
4. Provide transformers and rectifiers for each strike as required. Verify voltage with electrical contractor.

2.13 KEYSWITCHES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
   a. Schlage 650 series

2. Acceptable Manufacturers and Products:
   a. No Substitute

B. Requirements:

1. Provide key switches capable of being configured to momentary or maintained action.
2. Provide key switches that accept a mortise cylinder. Cylinders: Refer to "KEYING" article, herein.

2.14 POWER SUPPLIES

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
    a. Schlage/Von Duprin PS900 Series

2. Acceptable Manufacturers and Products:
    a. No Substitute

B. Requirements:

1. Provide power supplies approved by manufacturer of supplied electrified hardware.
2. Provide appropriate quantity of power supplies necessary for proper operation of electrified locking components as recommended by manufacturer of electrified locking components with consideration for each electrified component using power supply, location of power supply, and approved wiring diagrams. Locate power supplies as directed by Architect.
3. Provide regulated and filtered 24 VDC power supply, and UL class 2 listed.
4. Provide power supplies with the following features:
    a. 12/24 VDC Output, field selectable.
    b. Class 2 Rated power limited output.
    c. Universal 120-240 VAC input.
    d. Low voltage DC, regulated and filtered.
    e. Polarized connector for distribution boards.
    f. Fused primary input.
    g. AC input and DC output monitoring circuit w/LED indicators.
    h. Cover mounted AC Input indication.
    i. Tested and certified to meet UL294.
    j. NEMA 1 enclosure.
    k. Hinged cover w/lock down screws.
    l. High voltage protective cover.

2.15 CYLINDERS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
    a. Schlage Everest 29 Primus XP

2. Acceptable Manufacturers and Products:
    a. No Substitute

B. Requirements:

1. Provide cylinders/cores, compliant with ANSI/BHMA A156.5; latest revision; cylinder face finished to match lockset, manufacturer’s series as indicated. Refer to “KEYING” article, herein.
2. Provide cylinders in the below-listed configuration(s), distributed throughout the Project as indicated.
    a. High Security: dual-locking cylinder with permanent core requiring restricted, patented keyway. Dual-locking mechanism with interlocking finger pin(s) to check for patented features on keys.

3. Patent Protection: Cylinders/cores requiring use of restricted, patented keys, patent protected.
4. Nickel silver bottom pins.

2.16 CYLINDERS

A. Manufacturers:

1. Scheduled Manufacturer and Product:
   a. MATCH EXISTING SCHLAGE MASTERKEYED SYSTEM - TO BE FIELD VERIFIED

2. Acceptable Manufacturers and Products:
   a. No Substitute

B. Requirements:

1. Provide cylinders/cores to match Owner’s existing key system, compliant with ANSI/BHMA A156.5; latest revision; cylinder face finished to match lockset, manufacturer’s series as indicated. Refer to “KEYING” article, herein.

2.17 KEYING

A. Scheduled System:

1. Existing factory registered system:
   a. Provide cylinders/cores keyed into Owner’s existing factory registered keying system. Comply with guidelines in ANSI/BHMA A156.28, incorporating decisions made at keying conference.

B. Requirements:

1. Construction Keying:
   a. Replaceable Construction Cores.
      1) Provide temporary construction cores replaceable by permanent cores, furnished in accordance with the following requirements.
         a) 3 construction control keys
         b) 12 construction change (day) keys.
      2) Owner or Owner’s Representative will replace temporary construction cores with permanent cores.

2. Permanent Keying:
   a. Provide permanent cylinders/cores keyed by the manufacturer according to the following key system.
      1) Master Keying system as directed by the Owner.
   b. Forward bitting list and keys separately from cylinders, by means as directed by Owner. Failure to comply with forwarding requirements will be cause for replacement of cylinders/cores involved at no additional cost to Owner.
   c. Provide keys with the following features:
      1) Material: Nickel silver; minimum thickness of .107-inch (2.3mm)
      2) Patent Protection: Keys and blanks protected by one or more utility patent(s).
      3) Geographically Exclusive: Where High Security or Security cylinders/cores are indicated, provide nationwide, geographically exclusive key system complying with the following restrictions.
   d. Identification:
      1) Mark permanent cylinders/cores and keys with applicable blind code for identification. Do not provide blind code marks with actual key cuts.
      2) Identification stamping provisions must be approved by the Architect and Owner.

3) Stamp cylinders/cores and keys with Owner’s unique key system facility code as established by the manufacturer; key symbol and embossed or stamped with “DO NOT DUPLICATE” along with the “PATENTED” or patent number to enforce the patent protection.
4) Failure to comply with stamping requirements will be cause for replacement of keys involved at no additional cost to Owner.
5) Forward permanent cylinders/cores to Owner, separately from keys, by means as directed by Owner.

e. Quantity: Furnish in the following quantities.
1) Change (Day) Keys: 3 per cylinder/core.
2) Permanent Control Keys: 3.
3) Master Keys: 6.

2.18 KEY MANAGEMENT SOFTWARE

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product: Overtur Key System Management

B. Requirements

1. Provide a cloud based key management system that is SOC2 Type 2 Certified - cyber security compliance framework.
2. Provide system with ability of key system design, including:
a. Uploading and leveraging current plans in pdf format to determine function and traffic flow.
b. Uploading and utilizing a current door schedule using the Revit Plugin or in Excel format.
c. Leveraging a mobile app to survey existing conditions and upload data into project.
d. Collaborating with stakeholders, make notes, and view real time data.
e. Conducting key meetings using visual digital tools with capability to filter for keyed openings and applying as many key symbols as necessary to an opening.
f. Designing key system hierarchy and creating schematic, and visually referencing affected openings.
g. Creating key schedule with fully configured part numbers.

3. Provide training and onboarding, including:
a. Providing one-time personalized training by manufacturer via web conference to Owner or Owner identified key system personnel.
b. Access to software support team and self-service knowledge center.
c. Connecting Overtur to the factory master key software, enabling real time visibility of key system data.
d. Setting up a Company Admin to manage all user access and roles
e. Step by step, white glove assistance to set up your project and key system in Overtur, given the appropriate information is supplied.
f. Facilitate upload of key holder data and currently assigned keys

2.19 KEY CONTROL SYSTEM

A. Manufacturers:

1. Scheduled Manufacturer:
    a. Telkee

2. Acceptable Manufacturers:
    a. HPC
    b. Lund

B. Requirements:

1. Provide key control system, including envelopes, labels, tags with self-locking key clips, receipt forms, 3-way visible card index, temporary markers, permanent markers, and standard metal cabinet, all as recommended by system manufacturer, with capacity for 150% of number of locks required for Project.
    a. Provide complete cross index system set up by hardware supplier, and place keys on markers and hooks in cabinet as determined by final key schedule.
    b. Provide hinged-panel type cabinet for wall mounting.

2.20 DOOR CLOSERS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
    a. LCN 4040XP series

2. Acceptable Manufacturers and Products:
    a. No Substitute

B. Requirements:

1. Provide door closers conforming to ANSI/BHMA A156.4 Grade 1 requirements by BHMA certified independent testing laboratory. ISO 9000 certify closers. Stamp units with date of manufacture code.
2. Provide door closers with fully hydraulic, full rack and pinion action with high strength cast iron cylinder, and full complement bearings at shaft.
3. Cylinder Body: 1-1/2-inch (38 mm) diameter piston with 5/8-inch (16 mm) diameter double heat-treated pinion journal. QR code with a direct link to maintenance instructions.
4. Hydraulic Fluid: Fireproof, passing requirements of UL10C, and requiring no seasonal closer adjustment for temperatures ranging from 120 degrees F to -30 degrees F.
5. Spring Power: Continuously adjustable over full range of closer sizes, and providing reduced opening force as required by accessibility codes and standards. Provide snap-on cover clip, with plastic covers, that secures cover to spring tube.
6. Hydraulic Regulation: By tamper-proof, non-critical valves, with separate adjustment for latch speed, general speed, and backcheck. Provide graphically labelled instructions on the closer body adjacent to each adjustment valve. Provide positive stop on reg valve that prevents reg screw from being backed out.
7. Provide closers with solid forged steel main arms and factory assembled heavy-duty forged forearms for parallel arm closers.
8. Pressure Relief Valve (PRV) Technology: Not permitted.
9. Finish for Closer Cylinders, Arms, Adapter Plates, and Metal Covers: Powder coating finish which has been certified to exceed 100 hours salt spray testing as described in ANSI Standard A156.4 and ASTM B117, or has special rust inhibitor (SRI).

10. Provide special templates, drop plates, mounting brackets, or adapters for arms as required for details, overhead stops, and other door hardware items interfering with closer mounting.
11. Closers shall be capable of being upgraded by adding modular mechanical or electronic components in the field.

2.21 ELECTROMECHANICAL AUTOMATIC OPERATORS

A. Manufacturers and Products:

1. Scheduled Manufacturer and Product:
    a. LCN Senior Swing

2. Acceptable Manufacturers and Products:
    a. No Substitute

B. Requirements:

1. Provide low energy automatic operator units that are electromechanical design complying with ANSI/BHMA A156.19.
    a. Opening: Powered by DC motor working through reduction gears.
    b. Closing: Spring force.
    c. Manual, hydraulic, or chain drive closers: Not permitted.
    d. Operation: Motor is off when door is in closing mode. Door can be manually operated with power on or off without damage to operator. Provide variable adjustments, including opening and closing speed adjustment.
    e. Cover: Aluminum.

2. Provide units with manual off/auto/hold-open switch, push and go function to activate power operator, vestibule interface delay, electric lock delay, hold-open delay adjustable from 1 to 32 seconds, and logic terminal to interface with accessories, mats, and sensors.
3. Provide drop plates, brackets, and adapters for arms as required to suit details.
4. Provide motion sensors and/or actuator switches, and receivers for operation as specified. Provide weather-resistant actuators at exterior applications.
5. Provide key switches, with LED's, recommended and approved by manufacturer of automatic operator as required for function as described in operation description of hardware sets. Cylinders: Refer to "KEYING" article, herein.
6. Provide complete assemblies of controls, switches, power supplies, relays, and parts/material recommended and approved by manufacturer of automatic operator for each individual leaf. Actuators control both doors simultaneously at pairs. Sequence operation of exterior and vestibule doors with automatic operators to allow ingress or egress through both sets of openings as directed by Architect. Locate actuators, key switches, and other controls as directed by Architect.

2.22 DOOR TRIM

A. Manufacturers:

1. Scheduled Manufacturer:
    a. Ives

2. Acceptable Manufacturers:

      a. No Substitute

B. Requirements:

   1. Provide push plates, push bars, pull plates, pulls, and hands-free reversible door pulls with diameter and length as scheduled.

2.23 PROTECTION PLATES

A. Manufacturers:

   1. Scheduled Manufacturer:
      a. Ives

   2. Acceptable Manufacturers:
      a. No Substitute

B. Requirements:

   1. Provide protection plates with a minimum of 0.050 inch (1 mm) thick, beveled four edges as scheduled. Furnish with sheet metal or wood screws, finished to match plates.
   2. Size plates 1 1/2 inches (38 mm) less width of door on single doors, 2 inches (51 mm) less width of door on pairs of doors with a mullion, and doors with edge guards. Size plates 1 inch (25 mm) less width of door on pairs without a mullion or edge guards.
   3. At fire rated doors, provide protection plates over 16 inches high with UL label.

2.24 OVERHEAD STOPS AND OVERHEAD STOP/HOLDERS

A. Manufacturers:

   1. Scheduled Manufacturers:
      a. Glynn-Johnson

   2. Acceptable Manufacturers:
      a. No Substitute

B. Requirements:

   1. Provide overhead stop at any door where conditions do not allow for a wall stop or floor stop presents tripping hazard.
   2. Provide friction type at doors without closer and positive type at doors with closer.

2.25 DOOR STOPS AND HOLDERS

A. Manufacturers:

   1. Scheduled Manufacturer:
      a. Ives

   2. Acceptable Manufacturers:
      a. No Substitute

B. Provide door stops at each door leaf:

1. Provide wall stops wherever possible. Provide concave type where lockset has a push button of thumbturn.
2. Where wall stop cannot be used, provide medium duty surface overhead stop.
3. Provide roller bumper where doors open into each other and overhead stop cannot be used.

2.26 THRESHOLDS, SEALS, DOOR SWEEPS, AUTOMATIC DOOR BOTTOMS, AND GASKETING

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Zero International

2. Acceptable Manufacturers:
   a. No Substitute

B. Requirements:

1. Provide thresholds, weather-stripping, and gasketing systems as specified and per architectural details. Match finish of other items.
2. Smoke- and Draft-Control Door Assemblies: Where smoke- and draft-control door assemblies are required, provide door hardware that meets requirements of assemblies tested according to UL 1784 and installed in compliance with NFPA 105.
3. Provide door sweeps, seals, astragals, and auto door bottoms only of type where resilient or flexible seal strip is easily replaceable and readily available.
4. Size thresholds 1/2 inch (13 mm) high by 5 inches (127 mm) wide by door width unless otherwise specified in the hardware sets or detailed in the drawings.

2.27 SILENCERS

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Ives

2. Acceptable Manufacturers:
   a. No Substitute

B. Requirements:

1. Provide "push-in" type silencers for hollow metal or wood frames.
2. Provide one silencer per 30 inches (762 mm) of height on each single frame, and two for each pair frame.
3. Omit where gasketing is specified.

2.28 DOOR POSITION SWITCHES

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Schlage

2. Acceptable Manufacturers:
   a. No Substitute

B. Requirements:

1. Provide recessed or surface mounted type door position switches as specified.
2. Coordinate door and frame preparations with door and frame suppliers. If switches are being used with magnetic locking device, provide minimum of 4 inches (102 mm) between switch and magnetic locking device.

2.29 LATCH PROTECTORS

A. Manufacturers:

1. Scheduled Manufacturer:
   a. Ives

2. Acceptable Manufacturers:
   a. No Substitute

B. Provide stainless steel latch protectors of type required to function with specified lock.

2.30 FINISHES

A. FINISH: BHMA 626/652 (US26D); EXCEPT:

1. Hinges at Exterior Doors: BHMA 630 (US32D)
2. Aluminum Geared Continuous Hinges: BHMA 628 (US28)
3. Push Plates, Pulls, and Push Bars: BHMA 630 (US32D)
4. Protection Plates: BHMA 630 (US32D)
5. Overhead Stops and Holders: BHMA 630 (US32D)
6. Door Closers: Powder Coat to Match
7. Wall Stops: BHMA 630 (US32D)
8. Latch Protectors: BHMA 630 (US32D)
9. Weatherstripping: Clear Anodized Aluminum
10. Thresholds: Mill Finish Aluminum

PART 3 - EXECUTION

3.1 EXAMINATION

A. Field verify existing conditions and measurements at existing doors and frames receiving new hardware. Verify that new hardware is compatible with existing door and frame preparation and existing conditions.

1. Where existing wall conditions will not allow door to swing using the scheduled hinges, provide wide-throw hinges and if needed, extended arms on closers.

B. Prior to installation of hardware, examine doors and frames, with Installer present, for compliance with requirements for installation tolerances, labeled fire-rated door assembly construction, wall and floor construction, and other conditions affecting performance. Verify doors, frames, and walls have been properly reinforced for hardware installation.

C. Examine roughing-in for electrical power systems to verify actual locations of wiring connections before electrified door hardware installation.

D. Submit a list of deficiencies in writing and proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Mount door hardware units at heights to comply with the following, unless otherwise indicated or required to comply with governing regulations.

1. Standard Steel Doors and Frames: ANSI/SDI A250.8.
2. Custom Steel Doors and Frames: HMMA 831.
3. Interior Architectural Wood Flush Doors: ANSI/WDMA I.S. 1A
4. Installation Guide for Doors and Hardware: DHI TDH-007-20

B. Install door hardware in accordance with NFPA 80, NFPA 101 and provide post-install inspection, testing as specified in section 1.03.E unless otherwise required to comply with governing regulations.

C. Install each hardware item in compliance with manufacturer's instructions and recommendations, using only fasteners provided by manufacturer.

D. Do not install surface mounted items until finishes have been completed on substrate. Protect all installed hardware during painting.

E. Set units level, plumb and true to line and location. Adjust and reinforce attachment substrate as necessary for proper installation and operation.

F. Drill and countersink units that are not factory prepared for anchorage fasteners. Space fasteners and anchors according to industry standards.

G. Install operating parts so they move freely and smoothly without binding, sticking, or excessive clearance.

H. Hinges: Install types and in quantities indicated in door hardware schedule but not fewer than quantity recommended by manufacturer for application indicated or one hinge for every 30 inches (750 mm) of door height, whichever is more stringent, unless other equivalent means of support for door, such as spring hinges or pivots, are provided.

I. Intermediate Offset Pivots: Where offset pivots are indicated, provide intermediate offset pivots in quantities indicated in door hardware schedule but not fewer than one intermediate offset pivot per door and one additional intermediate offset pivot for every 30 inches (750 mm) of door height greater than 90 inches (2286 mm).

J. Lock Cylinders:

1. Install construction cores to secure building and areas during construction period.
2. Replace construction cores with permanent cores as indicated in keying section.
3. Furnish permanent cores to Owner for installation.

K. Wiring: Coordinate with Division 26, ELECTRICAL and Division 28 ELECTRONIC SAFETY AND SECURITY sections for:

1. Conduit, junction boxes and wire pulls.
2. Connections to and from power supplies to electrified hardware.
3. Connections to fire/smoke alarm system and smoke evacuation system.
4. Connection of wire to door position switches and wire runs to central room or area, as directed by Architect.
5. Connections to panel interface modules, controllers, and gateways.
6. Testing and labeling wires with Architect's opening number.

L. Key Control System: Tag keys and place them on markers and hooks in key control system cabinet, as determined by final keying schedule.

M. Door Closers: Mount closers on room side of corridor doors, inside of exterior doors, and stair side of stairway doors from corridors. Mount closers so they are not visible in corridors, lobbies and other public spaces unless approved by Architect.

N. Closer/Holders: Mount closer/holders on room side of corridor doors, inside of exterior doors, and stair side of stairway doors.

O. Kickplates: Mount kickplates and armor plates flush to bottom of door, centered on door leaf between jambs or jamb seals and mullions.

P. Power Supplies: Locate power supplies as indicated or, if not indicated, above accessible ceilings or in equipment room, or alternate location as directed by Architect.

Q. Thresholds: Set thresholds in full bed of sealant complying with requirements specified in Division 07 Section "Joint Sealants."

R. Stops: Provide wall stops for doors unless floor or other type stops are indicated in door hardware schedule. Do not mount floor stops where they may impede traffic or present tripping hazard.

S. Perimeter Gasketing: Apply to head and jamb, forming seal between door and frame.

T. Meeting Stile Gasketing: Fasten to meeting stiles, forming seal when doors are closed.

U. Door Bottoms and Sweeps: Apply to bottom of door, forming seal with threshold when door is closed.

3.3 ADJUSTING

A. Initial Adjustment: Adjust and check each operating item of door hardware and each door to ensure proper operation or function of every unit. Replace units that cannot be adjusted to operate as intended. Adjust door control devices to compensate for final operation of heating and ventilating equipment and to comply with referenced accessibility requirements.

1. Electric Strikes: Adjust horizontal and vertical alignment of keeper to properly engage lock bolt.
2. Door Closers: Adjust sweep period to comply with accessibility requirements and requirements of authorities having jurisdiction.

B. Occupancy Adjustment: Approximately three to six months after date of Substantial Completion, examine and readjust each item of door hardware, including adjusting operating forces, as necessary to ensure function of doors and door hardware.

3.4 CLEANING AND PROTECTION

A. Clean adjacent surfaces soiled by door hardware installation.

B. Clean operating items per manufacturer's instructions to restore proper function and finish.

C. Provide final protection and maintain conditions that ensure door hardware is without damage or deterioration at time of Substantial Completion.

3.5 DOOR HARDWARE SCHEDULE

A. Hardware items are referenced in the following hardware schedule. Refer to the above specifications for special features, options, cylinders/keying, and other requirements.

B. Hardware Sets:

140612 OPT0454670 Version 6

HARDWARE GROUP NO. 02

100A

PROVIDE EACH PR DOOR(S) WITH THE FOLLOWING:

2	EA	PIVOT SET	7215 SET	626	IVE
2	EA	INTERMEDIATE PIVOT	7215 INT	626	IVE
1	EA	REMOVABLE MULLION	KR4954 STAB	689	VON
1	EA	ELEC PANIC HARDWARE	RX-LC-QEL-99-EO W/CYL HOLE-CON-SNB 24 VDC	626	VON
1	EA	ELEC PANIC HARDWARE	RX-LC-QEL-99-EO-CON-SNB 24 VDC	626	VON
1	EA	MULLION STORAGE KIT	MT54	689	VON
1	EA	FSIC RIM CYLINDER	20-057 ICX W/CONST. CORE	626	SCH
2	EA	FSIC MORTISE CYLINDER	20-061 ICX W/CONST. CORE	626	SCH
3	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
2	EA	LONG DOOR PULL	9264F 36" 20" STD	630-316	IVE
1	EA	SURF. AUTO OPERATOR W/ MOUNTING PLATE	9553 LONG2 LESS TRACK HL/D AS REQ (120/240 VAC)	ANCLR	LCN
1	EA	ACTUATOR WITH BOLLARD KIT	8310-3836T	689	LCN
1	EA	ACTUATOR, TOUCHLESS	8310-813J - INTERIOR MULLION MOUNT		LCN
2	EA	ACTIVATION RECEIVER	8310-865 - AS REQUIRED		LCN
1	EA	MULLION SEAL	139N PSA		ZER
		SEALS - AS TESTED BY DOOR MANUFACTURER	BY DOOR / FRAME MANUFACTURER		
1	EA	THRESHOLD & 2 SWEEPS	THRESHOLD AND SWEEP by Door/Frame Mfgr.		B/O
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	KEY SWITCH	653-1414 L2 12/24 VDC - FOR AUTOMATIC OPERATOR	630	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	POWER SUPPLY	PS906 900-4RL 900-2RS 900-BBK 120/240 VAC		VON
		NOTE	FURNISH ADDITIONAL PIVOTS AS REQUIRED FOR DOORS 7'6" AND OVER		

OPERATIONAL DESCRIPTION: DURING REGULAR BUSINESS HRS DOORS TO BE UNLOCKED AND AUTO OPERATOR FUNCTION TO BE ENABLED. AUTO DOOR OPERATOR CONTROLLED BY INTERIOR WAVE ACTUATOR OR EXTERIOR BOLLARD ACTUATOR. AUTO DOOR OPERATOR TO SEQUENCE AND RETRACT ELECTRIC STRIKE ALLOWING DOOR TO OPEN. ALWAYS FREE EGRESS. FAIL SECURE. KEYSWITCH TO UNLOCK ELECTRIC STRIKE FOR BEST REPEATER. KEYSWITCH TO ALSO CONTROL ACTUATORS.
AFTER HRS OPERATION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. CREDENTIAL READER TO UNLOCK EXTERIOR ACTUATOR FOR AUTO DOOR OPERATOR. ACTUATORS ACTIVATE AUTO OPERATOR. AUTO OPERATOR TO SEQUENCE AND RETRACT ELECTRIC STRIKE
ALLOWING DOOR TO OPEN. ALWAYS FREE EGRESS. FAIL SECURE ELECTRIC STRIKES. OPTIONAL DOOR CONTROL BY SECURITY SYSTEM.

ADA OPERATION: SEQUENCE BOTH LHR LEAVES FOR ADA EXITING ON INTERIOR ACTUATOR ACTIVATION.

HARDWARE GROUP NO. 03

100B

PROVIDE EACH PR DOOR(S) WITH THE FOLLOWING:

2	EA	PIVOT SET	7215 SET	626	IVE
2	EA	INTERMEDIATE PIVOT	7215 INT	626	IVE
2	EA	DUMMY PUSH BAR	350	628	VON
2	EA	LONG DOOR PULL	9264F 36" 20" STD	630-316	IVE
1	EA	SURF. AUTO OPERATOR W/ MOUNTING PLATE	9553 LONG2 LESS TRACK HL/D AS REQ (120/240 VAC)	ANCLR	LCN
1	EA	ACTUATOR, TOUCHLESS	8310-813 - WALL MOUNT	BLK	LCN
1	EA	ACTUATOR, TOUCHLESS	8310-813J - MULLION MOUNT		LCN
1	SET	SEA	PERIMETER SEAL BY FRAME MANUFACTURER		
1	SET	ASTRAGAL	MEETING STILE SEAL BY DOOR MANUFACTURER		
		NOTE	FURNISH ADDITIONAL PIVOTS AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 05

136A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP	630	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626 RX	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	SURFACE CLOSER	4040XP SCUSH SRI TBSRT	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	RAIN DRIP - AS REQUIRED	142AA	AA	ZER
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
1	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

OPERATIONAL DESCRIPTION: DOOR MONITORED ONLY BY DOOR CONTACT. REQUEST TO EXIT SWITCH SIGNALS AUTHORIZED EGRESS.

HARDWARE GROUP NO. 06

108A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

1	EA	PIVOT SET	7215 SET	626	IVE
1	EA	INTERMEDIATE PIVOT	7215 INT	626	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	ELEC PANIC HARDWARE	LD-RX-LC-99-L-03-CON-SNB	626	VON
1	EA	FSIC RIM CYLINDER	20-057 ICX W/CONST. CORE	626	SCH
1	EA	ELECTRIC STRIKE	6300 FSE 12/24 VAC/VDC	630	VON
1	EA	LONG DOOR PULL	9264F 36" 20" STD	630-316	IVE
1	EA	SURFACE CLOSER	4040XP SCUSH SRI TBSRT	689	LCN
1	SET	SEA	PERIMETER SEAL BY FRAME MANUFACTURER		
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		DIAGRAMS	PROVIDE FACTORY POINT TO POINT WIRING DIAGRAMS		B/O
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

CARD ACCESS SYSTEM, MULTI-TECH READER, POWER SUPPLY, WIRING AND CONNECTIONS FURNISHED, INSTALLED AND COMMISSIONED BY DIVISION 28.

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. CREDENTIAL READER TO UNLATCH ELECTRIC STRIKE ALLOWING MANUAL ENTRY. ALWAYS FREE EGRESS. ELECTRIC STRIKE FAIL SECURE.

HARDWARE GROUP NO. 07

134A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP	630	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	ELEC PANIC HARDWARE	LD-RX-LC-99-L-03-CON-SNB	626	VON
1	EA	FSIC RIM CYLINDER	20-057 ICX W/CONST. CORE	626	SCH
1	EA	ELECTRIC STRIKE	6300 FSE 12/24 VAC/VDC	630	VON
1	EA	SURFACE CLOSER	4040XP SCUSH SRI TBSRT	689	LCN
1	EA	RAIN DRIP - AS REQUIRED	142AA	AA	ZER
1	SET	SEA	PERIMETER SEAL BY FRAME MANUFACTURER		
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		DIAGRAMS	PROVIDE FACTORY POINT TO POINT WIRING DIAGRAMS		B/O
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

CARD ACCESS SYSTEM, MULTI-TECH READER, POWER SUPPLY, WIRING AND CONNECTIONS FURNISHED, INSTALLED AND COMMISSIONED BY DIVISION 28.

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. CREDENTIAL READER TO UNLATCH ELECTRIC STRIKE ALLOWING MANUAL ENTRY. ALWAYS FREE EGRESS. ELECTRIC STRIKE FAIL SECURE.

HARDWARE GROUP NO. 08

108B

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

1	EA	PIVOT SET	7215 SET	626	IVE
1	EA	INTERMEDIATE PIVOT	7215 INT	626	IVE
1	EA	DUMMY PUSH BAR	350	628	VON
1	EA	LONG DOOR PULL	9264F 36" 20" STD	630-316	IVE
1	EA	SURFACE CLOSER	4040XP SCUSH TBSRT	689	LCN
1	EA	PA MOUNTING PLATE - AS REQUIRED	4040XP-18PA SRT	689	LCN
1	EA	CUSH SHOE SUPPORT - AS REQUIRED	4040XP-30 SRT	689	LCN
1	EA	BLADE STOP SPACER - AS REQUIRED	4040XP-61 SRT	689	LCN
1	SET	SEA	PERIMETER SEAL BY FRAME MANUFACTURER		
1	SET	ASTRAGAL	MEETING STILE SEAL BY DOOR MANUFACTURER		
		NOTE	FURNISH ADDITIONAL PIVOTS AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 10

E119C

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	630	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626 RX	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	ELECTRIC STRIKE	6211 FSE CON 12/16/24/28 VAC/VDC	630	VON
1	EA	SURFACE CLOSER	4040XP SRI TBSRT	689	LCN
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	RAIN DRIP - AS REQUIRED	142AA	AA	ZER
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		NOTE:	FURNISH 5BB1HW 5" X 4.5" HINGES AT DOORS OVER 3'0" WIDE		
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. ALWAYS FREE EGRESS. FAIL SECURE.

HARDWARE GROUP NO. 11

140A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	FIRE EXIT HARDWARE	99-L-F-03-SNB	626	VON
1	EA	FSIC RIM CYLINDER	20-057 ICX W/CONST. CORE	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	ARMOR PLATE	8402 34" X AS REQ B-CS TKTX	630	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
		NOTE:	FURNISH 5BB1HW 5" X 4.5" HINGES AT DOORS OVER 3'0" WIDE		
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 12

140B 149A 149C 149E 149H

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP	630	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	ELEC PANIC HARDWARE	LD-RX-LC-99-L-03-CON-SNB	626	VON
1	EA	FSIC RIM CYLINDER	20-057 ICX W/CONST. CORE	626	SCH
1	EA	ELECTRIC STRIKE	6300 FSE 12/24 VAC/VDC	630	VON
1	EA	SURFACE CLOSER	4040XP SCUSH SRI TBSRT	689	LCN
1	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
1	EA	RAIN DRIP - AS REQUIRED	142AA	AA	ZER
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		DIAGRAMS	PROVIDE FACTORY POINT TO POINT WIRING DIAGRAMS		B/O
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

CARD ACCESS SYSTEM, MULTI-TECH READER, POWER SUPPLY, WIRING AND CONNECTIONS FURNISHED, INSTALLED AND COMMISSIONED BY DIVISION 28.

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. CREDENTIAL READER TO UNLATCH ELECTRIC STRIKE ALLOWING MANUAL ENTRY. ALWAYS FREE EGRESS. ELECTRIC STRIKE FAIL SECURE.

HARDWARE GROUP NO. 18

E120A E121B

PROVIDE EACH UEP DOOR(S) WITH THE FOLLOWING:

6	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	CONST LATCHING BOLT - AS REQUIRED	FB52/FB62	630	IVE
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626 10-072 7/8" LIP	626	SCH
1	EA	FSIC RIM CYLINDER	20-057 ICX W/CONST. CORE	626	SCH
1	EA	ELECTRIC STRIKE - FAIL SECURE VOLTAGE AS REQUIRED - PAIR DOORS	6223 FSE CON 12/16/24/28 VAC/VDC	630	VON
1	EA	COORDINATOR	COR X FL X MB - AS REQUIRED	711	IVE
2	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	MOUNTING PLATE - AS REQUIRED	4040XP-18/18G/18TJ/18PA TBSRT - PROVIDE ON INACTIVE LEAF	689	LCN
2	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
2	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED - EXISTING READER TO BE RELOCATED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		DIAGRAMS	PROVIDE FACTORY POINT TO POINT WIRING DIAGRAMS		B/O

MULTI-TECH READER, POWER SUPPLY, WIRING AND CONNECTIONS FURNISHED, INSTALLED AND COMMISSIONED BY DIVISION 28.

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. CREDENTIAL READER TO UNLATCH ELECTRIC STRIKE ALLOWING MANUAL ENTRY. ALWAYS FREE EGRESS. ELECTRIC STRIKE FAIL SECURE.

HARDWARE GROUP NO. 19

104A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PANIC HARDWARE	CDSI-99-L-03-SNB	626	VON
1	EA	FSIC RIM CYLINDER	20-057 ICX W/CONST. CORE	626	SCH
1	EA	FSIC MORTISE CYLINDER	20-061 ICX W/CONST. CORE - CYLINDER DOGGING	626	SCH
2	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 21

105A	111A	113A	112A	114A	115A
116A	117A	118A	124A	129A	130A
131A	132A	133A	137A	E101A	E106A
E105A	E107A	E108A			

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1 4.5 X 4.5	652	IVE
1	EA	OFFICE/ENTRY LOCK	L9050T.03.626.A.626.03.626.A.626 09-544	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	WALL STOP	WS401/402CCV	626	IVE
3	EA	GASKETING	188S-BK PSA (OMIT @ NON-RATED DOORS, USE SILENCERS)	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 24

144A 145A 146A 147A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	630	IVE
1	EA	PRIVACY LOCK W/ OUTSIDE INDICATOR	L9040 03A.03.626.A.626.03.626.A.626 L583-363 OS-OCC	626	SCH
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
3	EA	GASKETING	188S-BK PSA (OMIT @ NON-RATED DOORS, USE SILENCERS)	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 26

109A 138A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 28

101A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP	630	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626 RX	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	ELECTRIC STRIKE	6211 FSE CON 12/16/24/28 VAC/VDC	630	VON
1	EA	LOCK GUARD	LG14	630	IVE
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	HD THRESHOLD	655A-V3-226 - 1/4" MAX CLEARANCE AT DOOR BOTTOM	A	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		NOTE:	FURNISH 5BB1HW 5" X 4.5" HINGES AT DOORS OVER 3'0" WIDE		
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

VERIFY ALL HARDWARE WITH BULLET RESISTANT DOOR MANUFACTURER:

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. ALWAYS FREE EGRESS. FAIL SECURE.

HARDWARE GROUP NO. 29

104C

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	CLASSROOM LOCK	L9070T.03.626.A.626.03.626.A.626	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	WALL STOP	WS401/402CCV	626	IVE
3	EA	GASKETING	188S-BK PSA (OMIT @ NON-RATED DOORS, USE SILENCERS)	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 31

125A 139A E125A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
3	EA	GASKETING	188S-BK PSA (OMIT @ NON-RATED DOORS, USE SILENCERS)	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 32

102A	103A	110A	126A	127A	128A
141A	E141A				

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PRIVACY LOCK W/ OUTSIDE INDICATOR	L9040 03A.03.626.A.626.03.626.A.626 L583-363 OS-OCC	626	SCH
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
		NOTE:	FURNISH 5BB1HW 5" X 4.5" HINGES AT DOORS OVER 3'0" WIDE		
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 33

E118B

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	OH STOP	90S	630	GLY
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	MOUNTING PLATE - AS REQUIRED	4040XP-18/18G/18PA TBSRT - AS REQUIRED	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	AUTO DOOR BOTTOM	369AA-Z49	AA	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 38

149B	149D	149F	149G	150A	150D

PROVIDE EACH RU DOOR(S) WITH THE FOLLOWING:

1	EA	FSIC MORTISE CYLINDER	20-061 ICX W/CONST. CORE	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	KEY SWITCH	653-0505 L2 36-079 12/24 VDC	630	SCE
		NOTE	REMAINDER OF HARDWARE BY DOOR MANUFACTURER		

COORDINATE LOCKING REQUIREMENTS WITH DOOR MANUFACTURER.

HARDWARE GROUP NO. 39

104B

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP	630	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	ELEC PANIC HARDWARE	LD-RX-LC-9975-L-03-CON-SNB	626	VON
1	EA	FSIC MORTISE CYLINDER	20-061 ICX W/CONST. CORE	626	SCH
1	EA	ELECTRIC STRIKE	6211 FSE CON 12/16/24/28 VAC/VDC	630	VON
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		DIAGRAMS	PROVIDE FACTORY POINT TO POINT WIRING DIAGRAMS		B/O
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

CARD ACCESS SYSTEM, MULTI-TECH READER, POWER SUPPLY, WIRING AND CONNECTIONS FURNISHED, INSTALLED AND COMMISSIONED BY DIVISION 28.

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. CREDENTIAL READER TO UNLATCH ELECTRIC STRIKE ALLOWING MANUAL ENTRY. ALWAYS FREE EGRESS. ELECTRIC STRIKE FAIL SECURE.

HARDWARE GROUP NO. 40

106A 120A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 41

148A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	OH STOP	100S	630	GLY
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 42

123A 135A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP/HOLDER	WS45/WS45X AS REQ	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 43

133B

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626 RX	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	ELECTRIC STRIKE	6211 FSE CON 12/16/24/28 VAC/VDC	630	VON
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		NOTE:	FURNISH 5BB1HW 5" X 4.5" HINGES AT DOORS OVER 3'0" WIDE		
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. ALWAYS FREE EGRESS. FAIL SECURE.

HARDWARE GROUP NO. 44

142B

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP	630	IVE
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	SURFACE CLOSER	4040XP SCUSH TBSRT	689	LCN
1	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 45

143A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 NRP	630	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	SURFACE CLOSER	4040XP SCUSH TBSRT	689	LCN
1	EA	KICK PLATE	8400 10" X AS REQ B-CS TKTX	630	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 46

150C E144A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	SURFACE CLOSER	4040XP SCUSH SRI TBSRT	689	LCN
1	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
3	EA	GASKETING	188S-BK PSA (OMIT @ NON-RATED DOORS, USE SILENCERS)	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 47

E118A E129A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
		NOTE:	BALANCE OF EXISTING HARDWARE TO REMAIN		B/O

REMOVE EXISTING CARD READER (DOOR E129A ONLY.) CHANGE LOCK TO PASSAGE.

HARDWARE GROUP NO. 48

E137C E139D

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	OH STOP	100S	630	GLY
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 49

E142A E143A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 50

E143B

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	OH STOP	100S	630	GLY
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 51

E119B

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	OH STOP & HOLDER	100F	630	GLY
1	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
3	EA	GASKETING	188S-BK PSA (OMIT @ NON-RATED DOORS, USE SILENCERS)	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 52

142A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
3	EA	GASKETING	188S-BK PSA (OMIT @ NON-RATED DOORS, USE SILENCERS)	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 53

121A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

1	EA	PIVOT SET	7215 SET	626	IVE
3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	652	IVE
1	EA	PASSAGE SET	L9010.03.626.A.626.03.626.A.626	626	SCH
1	EA	OH STOP	100S	630	GLY
1	EA	REG/ PA SURFACE CLOSER - AS REQUIRED	4040XP REG TBSRT/ 4040XP EDA TBSRT - AS REQUIRED	689	LCN
1	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

HARDWARE GROUP NO. 54

149J 150B

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	630	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626 RX	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	ELECTRIC STRIKE	6211 FSE CON 12/16/24/28 VAC/VDC	630	VON
1	EA	OH STOP	100S	630	GLY
1	EA	SURFACE CLOSER	4040XP SRI ST-1630 TBSRT	689	LCN
1	EA	TOP JAMB MTG PLATE	4040XP-18TJ SRI SRT	689	LCN
1	EA	RAIN DRIP - AS REQUIRED	142AA	AA	ZER
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		NOTE:	FURNISH 5BB1HW 5" X 4.5" HINGES AT DOORS OVER 3'0" WIDE		
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. ALWAYS FREE EGRESS. FAIL SECURE.

HARDWARE GROUP NO. 55

E114A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

1	EA	OFFICE/ENTRY LOCK	L9050T.03.626.A.626.03.626.A.626 09-544	626	SCH
		NOTE:	BALANCE OF EXISTING HARDWARE TO REMAIN		B/O

EXISTING DOOR AND EXISTING FRAME - EXCHANGE STOREROOM FUNCTION TO OFFICE, REUSE CYLINDER

HARDWARE GROUP NO. 56

E130A

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	630	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626 RX	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	ELECTRIC STRIKE	6211 FSE CON 12/16/24/28 VAC/VDC	630	VON
1	EA	SURFACE CLOSER	4040XP SRI TBSRT	689	LCN
1	EA	ARMOR PLATE	8400 34" X AS REQ B-CS TKTX	630	IVE
1	EA	WALL STOP	WS401/402CCV	626	IVE
1	EA	GASKETING SET	188SBK PSA	BK	ZER
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		NOTE:	FURNISH 5BB1HW 5" X 4.5" HINGES AT DOORS OVER 3'0" WIDE		
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. ALWAYS FREE EGRESS. FAIL SECURE.

HARDWARE GROUP NO. 57

E142D E139C

PROVIDE EACH SGL DOOR(S) WITH THE FOLLOWING:

3	EA	HINGE	5BB1HW 4.5 X 4.5 - NRP AT OUTSWINGING DRS	630	IVE
1	EA	POWER TRANSFER	EPT10 CON	689	VON
1	EA	STOREROOM LOCK	L9080T.03.626.A.626.03.626.A.626 RX	626	SCH
1	EA	FSIC PRIMUS PERMANENT CORE	20-740-XP CKC - MATCH EXISTING KEYWAY SYSTEM	626	SCH
1	EA	ELECTRIC STRIKE	6211 FSE CON 12/16/24/28 VAC/VDC	630	VON
1	EA	SURFACE CLOSER	4040XP SCUSH SRI TBSRT	689	LCN
1	EA	PA MOUNTING PLATE - AS REQUIRED	4040XP-18PA SRT	689	LCN
1	EA	CUSH SHOE SUPPORT - AS REQUIRED	4040XP-30 SRT	689	LCN
1	EA	BLADE STOP SPACER - AS REQUIRED	4040XP-61 SRT	689	LCN
1	EA	RAIN DRIP - AS REQUIRED	142AA	AA	ZER
1	SET	SEAL	PERIMETER SEAL BY FRAME MANUFACTURER		B/O
1	EA	DOOR SWEEP	39A	A	ZER
1	EA	HD THRESHOLD	655A-V3-226	A	ZER
1	EA	WIRE HARNESS - IN DOOR	CON - LENGTH AS REQUIRED - FROM LOCK TO EPT		SCH
2	EA	WIRE HARNESS CONNECTOR - IN FRAME	CON - 6W - FROM CON TO POWER SUPPLY		SCH
1	EA	MULTITECH READER - FURNISHED, INSTALLED, COMMISSIONED BY DIV 28	MT15B 5VDC - 28VDC - AS REQUIRED	BLK	SCE
1	EA	DOOR CONTACT - BY DIVISION 28	679-05 WD/HM - AS REQUIRED	BLK	SCE
1	EA	CREDENTIAL POWER SUPPLY - AS REQUIRED	BY DIVISION 28		B/O
		NOTE:	FURNISH 5BB1HW 5" X 4.5" HINGES AT DOORS OVER 3'0" WIDE		
		NOTE:	FURNISH ADDITIONAL HINGES AS REQUIRED FOR DOORS 7'6" AND OVER		

OPERATIONAL DESCRIPTION: ENTRANCE BY CREDENTIAL READER OR MANUAL KEY OVER-RIDE. ALWAYS FREE EGRESS. FAIL SECURE.

END OF SECTION